RAG Learning

1.3 Functions

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs f1–f5, factor theorem

What this topic covers. Polynomials and the factor theorem, then the language of functions — domain, range, composition, inverses — and the modulus function together with inequalities involving it.

1. Polynomials

A **polynomial** in x is a sum of terms $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ where the powers are non-negative integers. The **degree** is the highest power present.

Multiplying polynomials adds their degrees: degree 2 times degree 3 gives degree 5.

The factor theorem

If $f(a) = 0$ then $(x - a)$ is a **factor** of $f(x)$ — and conversely.

The closely related **remainder theorem** says the remainder on dividing $f(x)$ by $(x - a)$ is $f(a)$. A remainder of zero is exactly what "is a factor" means.

Worked example. Show that $(x + 3)$ is a factor of $f(x) = x^3 + 2x^2 - 5x - 6$, and factorise completely.
The factor $(x + 3)$ corresponds to $a = -3$:

$$f(-3) = -27 + 18 + 15 - 6 = 0$$

so $(x + 3)$ is a factor. Dividing gives

$$f(x) = (x+3)(x^2 - x - 2) = (x+3)(x-2)(x+1)$$

For $(x+3)$ you substitute $x = -3$, not $x = 3$. The sign slip here is extremely common — the factor $(x-a)$ means testing $+a$, so $(x+3) = (x-(-3))$ means testing -3.

Worked example. Given that $(x-2)$ is a factor of $x^3 + ax^2 - 5x + 6$, find a.

$$f(2) = 8 + 4a - 10 + 6 = 4 + 4a = 0 \implies a = -1$$

Worked example. Fully factorise $x^3 - 3x + 2$.
Try small values: $f(1) = 1 - 3 + 2 = 0$, so $(x-1)$ is a factor. Dividing gives $x^2 + x - 2 = (x+2)(x-1)$, so

$$x^3 - 3x + 2 = (x-1)^2(x+2)$$

a **repeated** root at $x = 1$ — the curve touches the axis there rather than crossing.

When hunting for a first factor, test the divisors of the constant term ($\pm1, \pm2, \pm3, \pm6$ for a constant of 6). Testing random values wastes time.

2. The language of functions

A **function** assigns exactly one output to each input. The **domain** is the set of permitted inputs; the **range** is the set of outputs actually produced.

Type	Meaning	Example
One-to-one	Each output comes from exactly one input	$f(x) = 2x + 1$
Many-to-one	Different inputs can share an output	$f(x) = x^2$

Restrictions on a domain usually come from two sources: you cannot divide by zero, and you cannot take the square root of a negative number.

Function	Natural domain	Range
$f(x) = x^2$	All real x	$f(x) \geq 0$
$f(x) = \sqrt{x-3}$	$x \geq 3$	$f(x) \geq 0$
$f(x) = \dfrac{1}{x}$	$x \neq 0$	$f(x) \neq 0$
$f(x) = \mathrm{e}^x$	All real x	$f(x) > 0$

Worked example. Find the range of $f(x) = x^2 - 4x + 7$ for real x.
Complete the square: $f(x) = (x-2)^2 + 3$. Since $(x-2)^2 \geq 0$, the least value is 3, so the range is $f(x) \geq 3$.

Completing the square is the standard route to the range of a quadratic. Reading the range off a rough sketch without it is where marks are lost.

3. Composite functions

$\mathrm{fg}(x)$ means $f\big(g(x)\big)$ — apply g **first**, then f.

Worked example. $f(x) = 2x + 3$ and $g(x) = x^2$. Find $\mathrm{fg}(x)$ and $\mathrm{gf}(x)$.

$$\mathrm{fg}(x) = f(x^2) = 2x^2 + 3$$

$$\mathrm{gf}(x) = g(2x+3) = (2x+3)^2$$

Composition is **not** commutative: $\mathrm{fg} \neq \mathrm{gf}$ in general, as above. The inner function is the one written next to the x.

A composite gf exists only if the range of f lies inside the domain of g. If f can output a negative number and g involves a square root, the composite fails for those inputs.

4. Inverse functions

The **inverse** f^{-1} reverses f: if $f(a) = b$ then $f^{-1}(b) = a$. An inverse exists precisely when f is **one-to-one**.

Method: write $y = f(x)$, rearrange to make x the subject, then swap the letters.

Worked example. Find the inverse of $f(x) = \dfrac{2x+1}{x-3}$.

$$y = \frac{2x+1}{x-3} \implies y(x-3) = 2x+1 \implies yx - 3y = 2x+1$$

$$x(y-2) = 3y+1 \implies x = \frac{3y+1}{y-2}$$

$$f^{-1}(x) = \frac{3x+1}{x-2}$$

Worked example. $f(x) = x^2 - 4x$ with domain $x \geq 2$. Find f^{-1}.
Complete the square: $y = (x-2)^2 - 4$, so $(x-2)^2 = y+4$ and $x - 2 = \pm\sqrt{y+4}$.

The domain $x \geq 2$ forces the **positive** root, so $f^{-1}(x) = 2 + \sqrt{x+4}$.

Restricting the domain is what makes a many-to-one function invertible — x^2 has no inverse over all the reals, but does for $x \geq 0$. When you take a square root during the rearrangement, use the given domain to decide the sign.

Two useful facts: the domain and range **swap** between f and f^{-1}, and the graph of $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in the line $y = x$. Consequently any intersection of a function with its inverse lies on $y = x$.

5. The modulus function

$|x|$ is the **magnitude** of x, ignoring sign:

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

The graph of $y = |x|$ is a V with its vertex at the origin. More generally $y = |x - a|$ has its vertex at $(a, 0)$.

Two graph transformations

Graph	How to construct it from $y = f(x)$
$y = \lvert f(x) \rvert$	Reflect the parts *below* the x-axis up into it. Nothing else moves.
$y = f(\lvert x \rvert)$	Keep the $x \geq 0$ part and reflect it in the y-axis, discarding the original $x < 0$ part.

These two are routinely confused. $|f(x)|$ acts on the **output**, so it folds the graph upwards. $f(|x|)$ acts on the **input**, so it makes the graph symmetric about the y-axis.

Modulus equations

Worked example. Solve $|2x - 1| = 5$.
Either $2x - 1 = 5$, giving $x = 3$; or $2x - 1 = -5$, giving $x = -2$.

Worked example. Solve $|x - 1| = |x + 3|$.
Squaring both sides (safe, as both are non-negative):

$$x^2 - 2x + 1 = x^2 + 6x + 9 \implies -8x = 8 \implies x = -1$$

Geometrically, $x = -1$ is the point equidistant from 1 and -3.

Modulus inequalities

Inequality	Equivalent to
$\lvert x \rvert < a$ (for $a > 0$)	$-a < x < a$
$\lvert x \rvert > a$ (for $a > 0$)	$x < -a$ or $x > a$

Worked example. Solve $|x + 2| > 3$.
Either $x + 2 > 3$, giving $x > 1$; or $x + 2 < -3$, giving $x < -5$. So $x < -5$ or $x > 1$.

Worked example. Solve $|2x - 3| < |x|$.
Both sides are non-negative, so squaring is valid:

$$4x^2 - 12x + 9 < x^2 \implies 3x^2 - 12x + 9 < 0 \implies x^2 - 4x + 3 < 0$$

$$(x - 1)(x - 3) < 0 \implies 1 < x < 3$$

Squaring is the reliable method when **both** sides are non-negative — which is automatic when both are moduli. It is *not* safe when one side could be negative, e.g. $|x| < x - 4$; there, split into cases instead.

6. Common pitfalls

- Testing $x = +3$ for the factor $(x + 3)$ instead of $x = -3$.
- Reading $\mathrm{fg}(x)$ as "f first" — it is g first.
- Claiming an inverse exists for a many-to-one function without restricting the domain.
- Taking the wrong sign of the square root when finding an inverse, ignoring the given domain.

- Swapping the effects of $|f(x)|$ and $f(|x|)$.
- Solving $|2x-1|=5$ with only the positive case, losing the second root.
- Squaring a modulus inequality where one side may be negative.
- Giving the range of a quadratic from a sketch rather than by completing the square.